

2D Modeling and Fuzzy Control of Slosh Dynamics in a Spacecraft

M. Navabi, A. Davoodi

Faculty of New Technologies Engineering
Shahid Beheshti University, GC.
davoodiali1369@gmail.com

Abstract—Fuel slosh dynamics can disturb attitude of spacecraft which has a partially filled container, in orbital maneuvering. So, it is necessary to control this harmful phenomenon. This paper models coupled dynamics of fuel sloshing and attitude of the spacecraft by two-pendulum model, and designs a fuzzy controller to control sloshing and attitude dynamics of the vehicle. Also, effectiveness of this controller is illustrated by conducted simulation.

Keywords—fuel sloshing; attitude control; orbital maneuver; fuzzy controller

I. INTRODUCTION

Controllers play essential role in scientific world such as stabilization of legged robots inspired biological control synergies based on orbital stability [1], linear differential inclusion [2] or robotic system for less-invasive treatment of osteolytic lesions behind an acetabular implant [3], and even they could be employed in daily human life such as stabilizing strings of vehicles to safety attain the maximum traffic flow [4]. Anyway, there are quite a few space and aerial missions such as formation flying [5-7], rendezvous in space [8], collision of space objects [9-11], missions in LEO [12-17], orbital maneuvering, simulating the earth magnetic field [18, 19], remote sensing satellites [20], docking maneuver of two spacecraft [21], path generation [22, 23] and systematic design [24] of aerial flying robots which are knowledge-based engineering. Attitude control is the most important problem in this category [25, 26], especially orbital maneuvering [27-29]. There are several innovative control methods which are employed for this area, for instance pulse width pulse frequency, pseudo-rate modulators [30], feedback linearization [31], adaptive feedback linearization [32], fuzzy [33] and integrated adaptive back stepping [34] methods. Also, attitude control is essential part in aerial vehicles designing [35-37]. The spacecraft which maneuvers in space needs plenty of fuel to introduce thrust force. Mass percentage of located fuel is high, so attitude of the spacecraft will be disturbed from desired position due to change in axial or angular acceleration [38]. This problem could bring heavy recompense, therefore it is necessary to control fuel sloshing dynamics. Interaction between attitude dynamics and fuel sloshing is a problem which

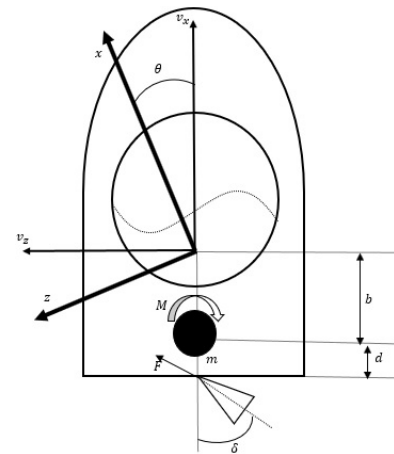


Fig. 1. Model of spacecraft and fuel container in 2D

has studied in notable number of papers. There are some studies which have proved that pendulum and mass-spring model can model fuel slosh dynamics in orbital transferring. Single pendulum is used in [39, 40], and single mass-spring is used in [40] to describe lowest frequency of fuel sloshing mode. Other studies used multi-pendulum [41-44] and mass-spring [45, 46] to characterize most fuel sloshing modes.

Current study present a two-pendulum model to characterize two modes of sloshing of fuel existing in spacecraft container. As seen in Fig. 1 the spacecraft is modeled by a rigid mass (m), and as seen in Fig. 2 fuel is modeled by two part: un-turbulent and sloshing fuel, and they are demonstrated by rigidly attached mass (m_0) and pendulum masses (m_i), respectively. After modeling, for controlling attitude of the spacecraft due to sloshing of fuel in orbital transfer a classic controller is proposed, but this controller doesn't have good performance. So, a fuzzy controller is designed to solve this problem. In this paper, gimbal deflection angle (δ) and torque about rigid mass (M) are control inputs. After designing controllers, effectiveness of them on dynamic parameters of the system is illustrated by a simulation.

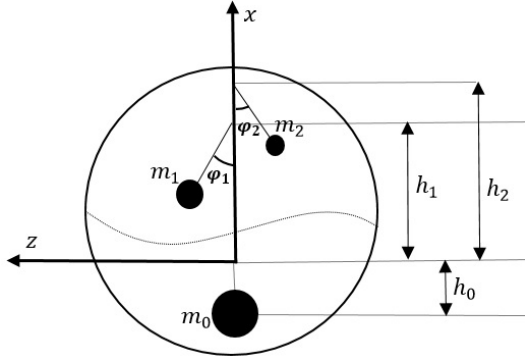


Fig. 2. Two-pendulum model of sloshing fuel

II. MODELING AND DERIVING EQUATIONS

Total mass of the spacecraft except fuel is denoted by a rigidly attached mass (m, I) which is located in center of the spacecraft mass, and fuel is modeled by a rigidly attached mass (m_0, I_0) which demonstrates un-turbulent fuel, and two pendulum masses $(m_i, l_i, i = 1, 2)$ which demonstrate sloshing fuel. Coupled equations of spacecraft and pendulum dynamics are derived by means of Lagrange equations.

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{V}} + \hat{\Omega} \times \frac{\partial L}{\partial V} = \tau_t \quad (1)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\Omega}} + \hat{\Omega} \times \frac{\partial L}{\partial \Omega} + \hat{V} \times \frac{\partial L}{\partial V} = \tau_r \quad (2)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\zeta}} - \frac{\partial L}{\partial \zeta} + \frac{\partial R}{\partial \dot{\zeta}} = 0 \quad (3)$$

Where $V = [v_x \ 0 \ v_z]^T$, $\Omega = [0 \ \Omega_2 \ 0]^T$ are vectors of axial and angular velocity of the rigid mass (m) , respectively. $\zeta = [\varphi_1 \ \varphi_2]^T$ is attitude angles vector of the pendulums, and τ_t is generalized forces vector, and τ_r is generalized torque which acts on center of spacecraft mass as well. Moreover, for a vector like x , $\hat{x}$ is a skew-symmetric matrix. L is Lagrangian and R is Rayleigh dissipation function, and they are written as:

$$L = \frac{1}{2} m (v_x^2 + (v_z + b\Omega)^2) + \frac{1}{2} m_0 (v_x^2 + (v_z + h_0\Omega)^2) + \frac{1}{2(I+I_0)\Omega^2} + \frac{1}{2} \sum_{i=1}^2 [m_i ((v_x + l_i(\Omega + \omega_i)\sin\varphi_i)^2 + (v_z - h_i\Omega + l_i(\Omega + \omega_i)\cos\varphi_i)^2) + I_i(\Omega + \omega_i)^2] \quad (4)$$

$$V = \begin{bmatrix} v_x \\ 0 \\ v_z \end{bmatrix} = \begin{bmatrix} \dot{x} \\ 0 \\ \dot{z} \end{bmatrix} + \hat{\Omega} \times \begin{bmatrix} x \\ 0 \\ z \end{bmatrix} \quad (5)$$

$$\zeta = \begin{bmatrix} \varphi_i \\ 0 \end{bmatrix}$$

$$R = \sum_{i=1}^2 \varepsilon_i \dot{\varphi}_i^2 \quad (6)$$

$$\tau_t = \begin{bmatrix} F\cos\delta \\ 0 \\ F\sin\delta \end{bmatrix}, \quad \tau_r = \begin{bmatrix} 0 \\ M + F(b+d)\sin\delta \\ 0 \end{bmatrix}$$

Where h_0, h_i are distances between fuel masses and center of container, and l_i is length of pendulum masses. Also, ε_i is damping coefficient of pendulum masses, and b is distance between center of container and center spacecraft mass, d is distance between center of spacecraft mass and gimbal pivot as well. By applying (1-3) coupled equations of the dynamic system can be derived as:

$$(m + m_0)a_x + mb\Omega^2 + \sum_{i=1}^2 [m_i a_x + m_i l_i (\dot{\Omega} + \ddot{\varphi}_i) \sin\varphi_i - m_i l_i \Omega^2 + m_i l_i (\Omega + \dot{\varphi}_i)^2 \cos\varphi_i] = F\cos\delta \quad (7)$$

$$(m + m_0)a_z + mb\dot{\Omega} + \sum_{i=1}^2 [m_i l_i (\dot{\Omega} + \ddot{\varphi}_i) \cos\varphi_i - m_i l_i \dot{\Omega} - m_i l_i (\Omega + \dot{\varphi}_i)^2 \sin\varphi_i] = F\sin\delta \quad (8)$$

$$(mb^2 + m_0 h_0^2) \dot{\Omega} + (I + I_0) \dot{\Omega} + mba_z - \sum_{i=1}^2 [m_i h_i^2 \dot{\Omega} + m_i l_i h_i ((\dot{\Omega} + \ddot{\varphi}_i) \cos\varphi_i - (\Omega + \dot{\varphi}_i)^2 \sin\varphi_i) - m_i l_i a_z - \varepsilon_i \dot{\varphi}_i] = M + F(b+d)\sin\delta \quad (9)$$

$$(I_i + m_i l_i^2) (\dot{\Omega} + \ddot{\varphi}_i) - m_i l_i h_i (\dot{\Omega} \cos\varphi_i + \Omega^2 \sin\varphi_i) + m_i l_i (a_x \sin\varphi_i + a_z \cos\varphi_i) + \varepsilon_i \dot{\varphi}_i = 0 \quad (10)$$

Also, a_x and a_z are accelerations of coordinates origin.

III. CONTROLLER DESIGNING

In this section, two controllers will be proposed to control attitude of the spacecraft and pendulums. By knowing the fact that it isn't necessary to control the axial velocity vector of the spacecraft along the x axis, first equation of the coupled equations is ignored. Now by changing the control inputs to new control inputs $(u_{1,2})$, coupled equations of motion will be rewritten as:

$$\dot{v}_z = u_1 + v_x \Omega \quad (11)$$

$$\dot{\Omega} = u_2 \quad (12)$$

$$\ddot{\varphi}_i = -\frac{m_i l_i}{I_i + m_i l_i^2} u_1 \cos\varphi_i - \frac{a_x m_i l_i}{I_i + m_i l_i^2} \sin\varphi_i - u_2 + \frac{m_i l_i h_i}{I_i + m_i l_i^2} (\cos\varphi_i u_2 + \Omega^2 \sin\varphi_i) - \frac{\varepsilon_i}{I_i + m_i l_i^2} \dot{\varphi}_i \quad (13)$$

These equations can be linearized about the origin value of v_z , θ , $\dot{\theta}$, φ_i , and $\dot{\varphi}_i$. The linearized equations are:

$$\dot{v}_z = u_1 + v_x \Omega \quad (14)$$

$$\dot{\Omega} = u_2 \quad (15)$$

$$\begin{aligned} \dot{\varphi}_i = & -\frac{m_i l_i}{I_i + m_i l_i^2} u_1 - \frac{a_x m_i l_i}{I_i + m_i l_i^2} \varphi_i - u_2 + \\ & \frac{m_i l_i h_i}{I_i + m_i l_i^2} u_2 - \frac{\varepsilon_i}{I_i + m_i l_i^2} \dot{\varphi}_i \end{aligned} \quad (16)$$

A. Classic Controller

New control inputs are introduced by proportional (P) and proportional-derivative (PD) theory and are written as:

$$u_1 = k_{p1} v_z \quad (17)$$

$$u_2 = k_{p2} \theta + k_{d2} \Omega \quad (18)$$

Where θ is attitude angle of the spacecraft.

B. Fuzzy Controller

The designed classic controller doesn't have good responsibility. So, a fuzzy controller is proposed to achieve control aims. For this purpose the inputs of classic controller are considered with some differences:

$$k_{p1} = k_{p1min} + (k_{p1max} - k_{p1min})k'_{p1} \quad (19)$$

$$k_{p2} = k_{p2min} + (k_{p2max} - k_{p2min})k'_{p2} \quad (20)$$

$$k_{d2} = k_{d2min} + (k_{d2max} - k_{d2min})k'_{d2} \quad (21)$$

Where

$$k_{p1min} = 0.64k_{p1}, \quad k_{p1max} = 1.2k_{p1}$$

$$k_{p2min} = 0.53k_{p2}, \quad k_{p2max} = k_{p2}, \quad k_{d2min} = k_{d2}, \quad k_{d2max} = 2k_{d2}$$

Also, k'_{pi} , k'_{d2} are Mamdani logic outputs, and as shown in Fig. 3 there are two Gaussian membership functions defining these outputs. Also, v_z , θ and Ω are inputs of this logic, and as illustrated in Fig. 4 they have five triangle and two trapeze membership functions.

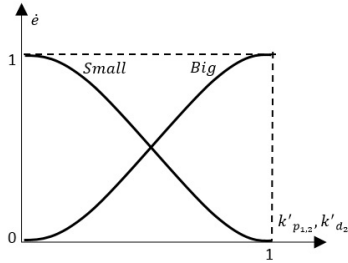


Fig. 3. Outputs' membership functions

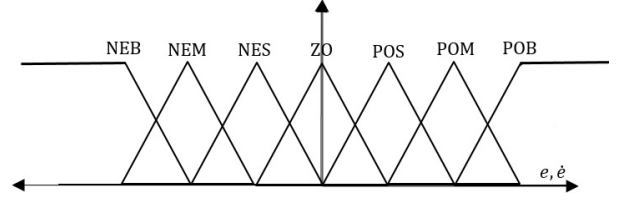


Fig. 4. Inputs' membership functions

The rules for this logic are defined using Tab. I as well as Tab. II [47]. Whereas, for defining the rules between k_{p1} and v_z , if v_z is NEB or POB, then k_{p1} is Big, and if v_z is NEM, NES, ZO, POS, or POM then k_{p1} is Small.

TABLE I. RULES FOR DEFINING k'_{p2}

$\dot{e}$		NEB	NEM	NES	ZO	POS	POM	POB
e	NEB	Big	Big	Big	Big	Big	Big	Big
	NEM	Small	Big	Big	Big	Big	Big	Small
	NES	Small	Small	Big	Big	Big	Small	Small
	ZO	Small	Small	Small	Big	Small	Small	Small
	POS	Small	Small	Big	Big	Big	Small	Small
	POM	Small	Big	Big	Big	Big	Big	Small
	POB	Big	Big	Big	Big	Big	Big	Big

TABLE II. RULES FOR DEFINING k'_{d2}

$\dot{e}$		NEB	NEM	NES	ZO	POS	POM	POB
e	NEB	Small	Small	Small	Small	Small	Small	Small
	NEM	Big	Big	Small	Small	Small	Big	Big
	NES	Big	Big	Big	Small	Big	Big	Big
	ZO	Big	Big	Big	Big	Big	Big	Big
	POS	Big	Big	Big	Small	Big	Big	Big
	POM	Big	Big	Small	Small	Small	Big	Big
	POB	Small	Small	Small	Small	Small	Small	Small

IV. SIMULATION

In this section to illustrate effectiveness of designed controllers, a simulation is conducted. The control aims are stabilizing attitude of the spacecraft and pendulums dynamic variables can be shown. As seen in Fig. 5 transverse velocity of the spacecraft after elapse 1400 seconds trend to zero, and it's obvious that performance of the fuzzy controller is better than the designed classic controller. Moreover, as shown in Figs. 6-7 attitude angle of the spacecraft and its angular velocity are becoming stable after 1000 seconds. Also, time behaviors of attitude angle of the pendulums and their angular velocity are shown in Figs. 8-9, and as these figures illustrate attitude angle after 1000 seconds trend to zero and angular velocity became zero almost at the beginning seconds. Time behaviors of control inputs are illustrated in Fig. 10.

So, by applying the designed controllers on linearized form of coupled equations of the dynamic system of spacecraft and pendulums, time behaviors of the spacecraft and pendulums dynamic variables can be shown. As seen in Fig. 5 transverse velocity of the spacecraft after elapse 1400 seconds trend to zero, and it's obvious that performance of the fuzzy controller is better than the designed classic controller. Moreover, as shown in Figs. 6-7 attitude angle of the spacecraft and its angular velocity are becoming stable after 1000 seconds. Also, time behaviors of attitude angle of the pendulums and their angular velocity are shown in Figs. 8-9, and as these figures illustrate attitude angle after 1000 seconds trend to zero and angular velocity became zero almost at the beginning seconds. Time behaviors of control inputs are illustrated in Fig. 10.

TABLE III. CONTROLLER GAINS

$k_{p1}=6 \times 10^{-3}$	
$k_{p2}=5 \times 10^{-5}$	$k_{d2}=6 \times 10^{-3}$

TABLE IV. PHYSICAL PARAMETERS OF THE SYSTEM

Parameter	Value	Parameter	Value
m	590 Kg	h_0	0.05 m
I	400 Kg.m ²	h_1	0.6 m
m_0	480 Kg	h_2	0.9 m
I_0	75 Kg.m ²	F	2250 N
m_1	50 Kg	l_1	0.2 m
l_1	10 Kg.m ²	l_2	0.1 m
m_2	5 Kg	ε_1	3.7 Kg.m ² /s
l_2	1 Kg.m ²	ε_2	0.5 Kg.m ² /s

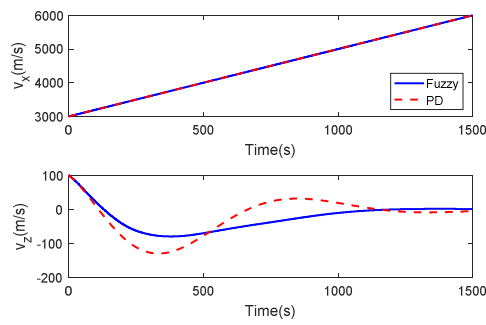


Fig. 5. Velocity vectors of the space robot

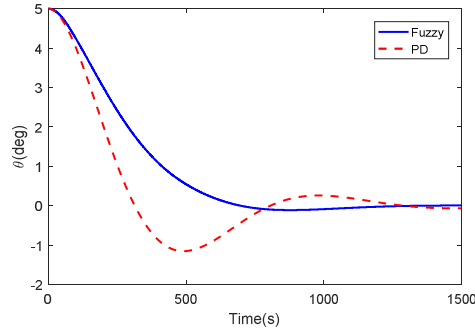


Fig. 6. Attitude angle of the space robot

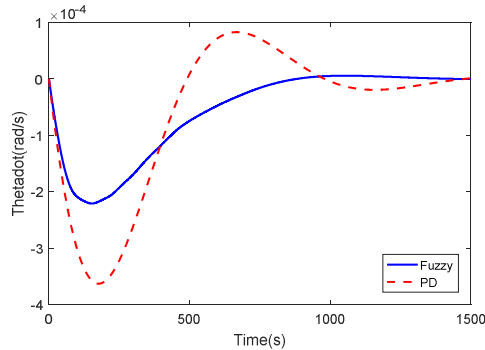


Fig. 7. Angular velocity of the space robot

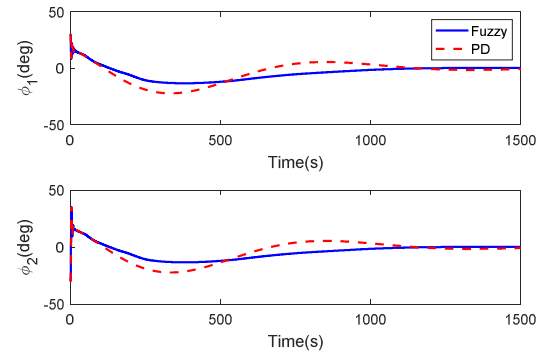


Fig. 8. Attitude angle of the pendulums

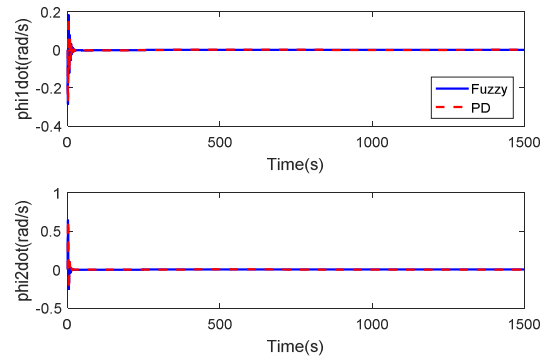


Fig. 9. Angular velocity of the pendulums

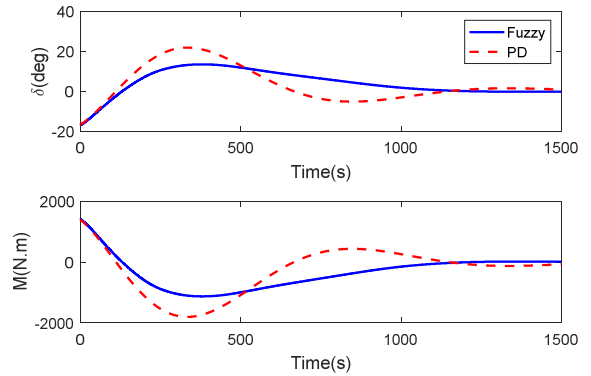


Fig. 10. Control inputs

V. CONCLUSION

Using two-pendulum model in two dimension, a complete nonlinear model of coupled dynamic system of attitude of the spacecraft and fuel sloshing has been developed, also a classic controller as well as fuzzy controller have been designed to stabilize attitude of the spacecraft and pendulums. The effectiveness of designed controllers has been illustrated by simulation, and as results show, although both of the controllers have good performance to achieve control aims, the fuzzy

controller stabilize dynamic parameters of the system with less overshoot in less time.

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